

Partitioning and Bucketing

```
> latitude,
> longitude,
> location,
> year AS crime_year
> FROM crimes_2022;
Query ID = hadoop_20260509062918_aef5924a-58aa-4898-b7c2-23e5a3371953
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1778304679972_0003)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	2	2	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0

```
VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 28.63 s
Loading data to table default.crimes_partitioned partition (crime_year=null)
Loaded : 1/1 partitions.
Time taken to load dynamic partitions: 0.519 seconds
Time taken for adding to write entity : 0.002 seconds
OK
```

Count total number of crimes

```
hive> SELECT COUNT(*) AS total_crime_records FROM crimes_2022;
Query ID = hadoop_20260509063106_914bf165-75dd-4b09-96f2-0790dc5ed5f7
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778304679972_0003)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	2	2	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0

```
VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 10.62 s
OK
473480
Time taken: 11.795 seconds, Fetched: 1 row(s)
```

Top 10 Crimes

```
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778304679972_0004)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	2	2	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0	0

```
VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 13.55 s
```

OK

NULL 232072

"THEFT" 54900

"BATTERY" 40970

"CRIMINAL DAMAGE" 27250

"MOTOR VEHICLE THEFT" 21473

```
Time taken: 12.828 seconds, Fetched: 1 row(s)
```

```
hive> SELECT
```

```
> crime_year,
> COUNT(*) AS total_crimes
> FROM crimes_partitioned
> GROUP BY crime_year
> ORDER BY crime_year;
```

```
Query ID = hadoop_20260509064354_775249c7-1b35-4746-b48a-654188515e2e
```

```
Total jobs = 1
```

```
Launching Job 1 out of 1
```

```
Status: Running (Executing on YARN cluster with App id application_1778304679972_0004)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 2	container	SUCCEEDED	2	2	0	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0	0

```
VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 6.94 s
```

OK

NULL 473480

```
Time taken: 7.805 seconds, Fetched: 1 row(s)
```